

Split/Merge AMR and Dynamic-Hydro Changes

EbF/improve_split_merge_amr_mesh

Audited 2026-08-26 at b0d4e8bff against origin/main (f760b022f).

This summary compares the branch with origin/main. Red gives the actual replaced equation or code path; green gives the branch implementation. Every user-facing control is stated with its default and enabling value. Changes are classified as features, MESA bug fixes, or development/test changes.

RED: replaced **GREEN: added/current** **BLUE: control or activation**

Features

Optional physics, numerical methods, and user-facing controls. Defaults preserve the previous behavior unless stated otherwise.

1. Weighted split/merge metric

FEATURE — U_FLAG / AMR

CHANGELOG: YES

PREVIOUS IMPLEMENTATION

Split/merge AMR selected one coordinate x at a time: r , $\ln r$, $\ln \tau$, or one of the hybrid radius–optical-depth coordinates. For the selected coordinate,

$$\Delta x_k = x_{R,k} - x_{L,k}, \quad \Delta x_* = \frac{x_{\max} - x_{\min}}{N_{\text{base}}}, \quad R_{\text{split}} = \frac{\Delta x_k}{\Delta x_*}.$$

The merge score used the selected-coordinate ratio and the mass floor, not the sum of all active coordinate components. A merge could therefore recreate a cell that immediately satisfied a different split requirement.

CURRENT IMPLEMENTATION

The opt-in metric normalizes and adds the active components,

$$\Delta \xi_k = w_R \frac{|\Delta \ln r_k|}{\Delta \ln r_*} + w_\tau \frac{|\Delta \ln \tau_k|}{\Delta \ln \tau_*}, \quad \Delta \xi_* = \frac{w_R + w_\tau}{N_{\text{base}}}.$$

The pairwise merge guard evaluates the same summed metric as the split test, so a permitted merge cannot immediately reproduce an oversized metric cell.

CONTROL / ACTIVATION

`split_merge_amr_use_metric_zoning_for_u_flag=.false.` by default; set it to `.true.`. Requires `use_split_merge_amr=.true.`, `u_flag=.true.`, and a positive `split_merge_amr_MaxLong`. Component weights are `split_merge_amr_metric_logR_weight=1d0` and `..._logtau_weight=1d0`; a weight > 0 enables that component. Normalization floors are `..._min_delta_lnr=1d-9` and `..._min_delta_lntau=1d-9`. There is no separate merge-guard control: every prospective metric merge is rejected when its summed metric exceeds `split_merge_amr_MaxLong` times the target size.

WHY

Resolve geometric and photospheric structure simultaneously without a systematic merge–split cycle.

PRIMARY CODE

`star/private/adjust_mesh_split_merge.f90`; `star/defaults/controls_dev.defaults`.

2. Bounded pressure reconstruction for split children

FEATURE — U_FLAG / AMR

CHANGELOG: YES

PREVIOUS IMPLEMENTATION

After a conservative split, each child received its reconstructed density and internal energy and was closed independently by the EOS. Pressure was only an EOS output:

$$P_{R,0} = P_{\text{EOS}}(\rho_R, e_R, X_R), \quad P_{L,0} = P_{\text{EOS}}(\rho_L, e_L, X_L).$$

No equation required $P_L - P_R$ to reproduce the limited parent-stencil pressure gradient; MLT turbulent pressure was not included in a target.

CURRENT IMPLEMENTATION

After the conservative velocity/energy split and radius correction, a bracketed EOS solve transfers internal energy between the children:

$$\Delta P_{\text{target}} = -\frac{1}{2} \nabla (P_{\text{EOS}} + P_{\text{turb}}) \Delta r_{\text{parent}},$$

$$e_R = e_{R,0} + \delta, \quad e_L = \frac{E_{\text{int,total}} - dm_R e_R}{dm_L}.$$

The accepted solution preserves mass, composition, momentum, and internal-plus-kinetic energy. Energy, pressure, and temperature remain inside their original stencil bounds; invalid, unbracketed, or unconverged EOS solves retain the ordinary conservative split.

CONTROL / ACTIVATION

`split_merge_amr_reconstruct_pressure_for_u_flag=.false.` by default; set it to `.true.`. Requires `use_split_merge_amr=.true.` and `u_flag=.true.`.

WHY

Reduce remesh-generated force imbalance without weakening any conservation invariant.

PRIMARY CODE

`star/private/adjust_mesh_split_merge.f90`; `star/defaults/controls_dev.defaults`.

3. Harmonic MLT/TDC dissipation length

FEATURE — MLT / TDC

CHANGELOG: NO

PREVIOUS IMPLEMENTATION

MLT and TDC used only the pressure-scale-height length

$$\Lambda_0 = \alpha_{\text{MLT}} H_P.$$

In weak-gravity layers, $H_P = P/(\rho g)$ can exceed the available radial geometry, and different TDC consumers could reconstruct the scale height separately.

CURRENT IMPLEMENTATION

The optional Wuchterl–Feuchtinger harmonic length (1998, A&A, 340, 419, Eq. 17) is

$$\frac{1}{\Lambda} = \frac{1}{\alpha_{\text{MLT}} H_P} + \frac{1}{\text{harmonic_dissipation_length_beta } r}.$$

The resulting Λ is used consistently in MLT/TDC source, damping, flux, diffusion, stored length, derived velocity, and eddy-viscous terms.

CONTROL / ACTIVATION

`harmonic_dissipation_length_beta=0d0` by default. A positive value enables the radial harmonic limiter; a nonpositive value gives exactly $\Lambda = \alpha_{\text{MLT}} H_P$. A positive value takes precedence when the mixing length is constructed, including over the older alternate-scale-height choice.

WHY

Limit the eddy scale smoothly by local radial geometry and make all TDC consumers use the same length.

PRIMARY CODE

`turb/private/mlt.f90`; `turb/private/tdc_support.f90`; `star/private/tdc_hydro.f90`.

4. Configurable EOS density floor

FEATURE — EOS

CHANGELOG: NO

PREVIOUS IMPLEMENTATION

The EOS support code rejected every call satisfying

$$\log_{10} \rho < -25$$

with a hard-coded threshold that could not be changed from an inlist.

CURRENT IMPLEMENTATION

The rejection test is

$$\log_{10} \rho < \text{min_logRho_for_eos},$$

with a lower permitted control bound of -30 .

CONTROL / ACTIVATION

`min_logRho_for_eos=-30d0`. This is an always-active threshold, not a Boolean toggle. Values below $-30d0$ are rejected while controls are read.

WHY

Permit lower-density dynamic ejecta while retaining a bound that protects high-order density derivatives from overflow.

PRIMARY CODE

star/private/eos_support.f90; star/defaults/controls.defaults.

5. Radiation-pressure floor for the momentum boundary

FEATURE — HYDRO BOUNDARY

CHANGELOG: YES

PREVIOUS IMPLEMENTATION

With the momentum outer boundary, the traction always used the atmosphere result,

$$P_{\text{momentum}} = P_{\text{atm}}, \quad P_{\text{gas,implied}} = P_{\text{atm}} - \frac{aT_{\text{bc}}^4}{3}.$$

A super-Eddington static atmosphere could therefore imply $P_{\text{gas}} < 0$.

CURRENT IMPLEMENTATION

The optional closure uses

$$P_{\text{momentum}} = \max\left(P_{\text{atm}}, \frac{aT_{\text{bc}}^4}{3}\right),$$

including the automatic-differentiation derivatives of the selected branch. The atmospheric temperature equation and other boundary choices are unchanged.

CONTROL / ACTIVATION

floor_momentum_outer_BC_at_Prad=.false. by default; set it to .true.. It is used only when use_momentum_outer_BC=.true..

WHY

Provide a nonnegative-gas-pressure traction when no static atmosphere solution exists and avoid dividing a missing surface pressure by a very small surface-cell mass.

PRIMARY CODE

star/private/hydro_eqns.f90; star/defaults/controls.defaults.

6. Constant-luminosity equation for idealized hydro

FEATURE — HYDRO EQUATIONS

CHANGELOG: YES

PREVIOUS IMPLEMENTATION

Every hydro model used the active thermal-transport residual (the ordinary gradient equation or the dP_{rad}/dm form) and the atmospheric surface-temperature equation. There was no equation imposing $L_k = L_{k+1}$, even for a benchmark intended to have prescribed constant luminosity.

CURRENT IMPLEMENTATION

The optional residual is

$$R_L(k) = \frac{L(k) - L(k+1)}{L_{\text{scale}}(k)}, \quad L(n_z + 1) = L_{\text{center}},$$

$$L_{\text{scale}}(k) = \max(1, |L_{\text{center}}|, |L_{\text{start}}(k)|, |L_{\text{next,start}}(k)|).$$

The same luminosity equation replaces the surface temperature boundary; the roundoff-sensitive delta_lgL limit is skipped for prescribed zero luminosity.

CONTROL / ACTIVATION

constant_L=.false. by default; set it to .true.. This selects the luminosity residual only; it does not select a momentum boundary or alter model-file storage.

WHY

Give Noh, Sedov, and similar tests their intended thermal closure without invoking radiative atmosphere physics.

PRIMARY CODE

star/private/hydro_temperature.f90; star/private/hydro_eqns.f90.

7. Composition-only Riemann-shock mixing reduction

FEATURE — U_FLAG / RIEMANN / MIXING

CHANGELOG: NO

PREVIOUS IMPLEMENTATION

The final symmetric chemical diffusion coefficient crossed a compressive HLLC shock unchanged. The available PPISN workaround acted earlier and more broadly: when velocity or velocity-jump ratios exceeded `max_v_div_cs_for_convection` or `max_abs_du_div_cs_for_convection`, it suppressed the convection state itself, including TDC velocity, flux, turbulent pressure, and eddy viscosity.

CURRENT IMPLEMENTATION

For inner state $L = k$ and outer state $R = k - 1$,

$$C_k = \max\left(0, \frac{u_L - u_R}{c_{s, \text{face}}}\right), \quad J_k = \max\left(0, \frac{P_*}{\min(P_L, P_R)} - 1\right), \quad S_k = \min(C_k, J_k).$$

Between S_{on} and S_{full} , $x = (S - S_{\text{on}})/(S_{\text{full}} - S_{\text{on}})$ and

$$f_D = 1 - x^3(10 - 15x + 6x^2).$$

Only final `D_mix`, `D_mix_non_rotation`, optional rotational mixing, and `cdc` are multiplied by f_D . The MLT/TDC dynamical state is unchanged.

CONTROL / ACTIVATION

Disabled by `Riemann_shock_D_mix_reduction_full_on=-1d0`. Enable with `...full_on>0d0`; require $0 \leq \dots_{\text{on}} < \dots_{\text{full_on}}$. Defaults are `..._on=0d0`, `..._full_on=-1d0`. Requires `u_flag=.true.`

WHY

Block symmetric abundance transport through a resolved shock without opening a one-face hole in TDC stress or heat flux.

PRIMARY CODE

`star/private/hydro_riemann.f90`; `star/private/mix_info.f90`.

8. Implicit and normalized entropy relaxation

FEATURE — RELAXATION / STAR_JOB

CHANGELOG: NO

PREVIOUS IMPLEMENTATION

Entropy relaxation always installed the explicit `other_energy` hook and used

$$\epsilon_{\text{old}} = \frac{e}{\tau} \left(1 - \frac{s}{s_{\text{target}}}\right).$$

Thus

$$\frac{ds}{dt} = \frac{e}{T\tau} \left(1 - \frac{s}{s_{\text{target}}}\right),$$

which is not an entropy e-folding law. Because the source was explicit, its local $\partial\epsilon/\partial\rho$ and $\partial\epsilon/\partial T$ terms were absent from the structure Jacobian. Relaxation also used the model's ordinary correction scale and retry hold for the entire interval.

CURRENT IMPLEMENTATION

The implicit-source option installs `use_other_energy_implicit` and evaluates T , s , and e as automatic-differentiation variables, so the local density and temperature derivatives enter the Newton matrix. Independently, the normalized-source option selects

$$\epsilon_{\text{norm}} = \frac{T(s_{\text{target}} - s)}{\tau}, \quad \frac{ds}{dt} = \frac{s_{\text{target}} - s}{\tau}.$$

The implicit option changes the hook/Jacobian path; it can be used with either the old energy-scaled formula or the normalized formula. All original hooks and solver settings are restored afterward.

CONTROL / ACTIVATION

Prerequisite: `relax_initial_entropy=.true.` In `&star_job`: `relax_entropy_use_implicit_source=.false.` and `relax_entropy_use_normalized_source=.false.`; set either independently to `.true.` `relax_entropy_scale_max_correction=0d0` means no temporary override; a positive value overrides it only during relaxation. `relax_entropy_retry_hold=-1` means no override; a nonnegative value overrides it only during relaxation.

WHY

Supply the requested entropy timescale and a genuinely implicit stiff source without permanently changing the model's solver policy.

PRIMARY CODE

star/private/relax.f90; star/defaults/star_job.defaults.

9. Turnover-time limiter for superadiabatic reduction

FEATURE — MLT / TDC / RESTART STATE

CHANGELOG: YES

PREVIOUS IMPLEMENTATION

The superadiabatic-reduction factor was applied instantaneously on every MLT or TDC evaluation. Writing $\eta = 1/\Gamma_{\text{factor}}$, the accepted value was simply

$$\eta^{n+1} = \eta_{\text{inst}},$$

with no response time and no accepted-step state to remap or save.

CURRENT IMPLEMENTATION

The optional limiter advances the applied multiplier toward the instantaneous value,

$$\eta^{n+1} = \eta^n + f_{\text{turn}}(\eta_{\text{inst}} - \eta^n),$$

where

$$f_{\text{turn}} = \min(\Delta t / \tau_{\text{conv}}, 1) \quad \text{or} \quad f_{\text{turn}} = 1 - \exp(-\Delta t / \tau_{\text{conv}}).$$

When a lagged convective velocity is available, $\tau_{\text{conv}} = H_P / \max(v_c, 10^{-10} \text{ cm s}^{-1})$; otherwise the Brunt timescale is used. The start-of-step timescale is fixed during Newton iterations. The accepted factor is preserved across retries, remeshing, surface cuts, and current-format photos.

CONTROL / ACTIVATION

superad_reduction_use_turnover_limit=.false. by default; set it to .true. together with use_superad_reduction=.true.. superad_reduction_turnover_limit_function='linear' by default; 'exponential' selects the exponential response.

WHY

Prevent the artificial gradient reduction from changing faster than the local convective response while keeping the choice fixed within one nonlinear solve.

PRIMARY CODE

star/private/turb_support.f90; star/private/turb_info.f90; star/private/star_utils.f90; photo and mesh state handling.

10. Temperature cutoff for superadiabatic reduction

FEATURE — MLT / TDC

CHANGELOG: YES

PREVIOUS IMPLEMENTATION

When use_superad_reduction was true, every eligible MLT or TDC face could be reduced regardless of temperature.

CURRENT IMPLEMENTATION

For a real model face ($k > 0$), reduction is evaluated only when

$$\log_{10}(T_{\text{start}}/\text{K}) < \text{superad_reduction_max_logT}.$$

At or above the cutoff, $\Gamma_{\text{factor}} = 1$ and the ordinary radiative gradient is retained. Using T_start prevents the active set from changing during Newton iterations. The $k = 0$ model-construction path is unchanged.

CONTROL / ACTIVATION

superad_reduction_max_logT=7d0 by default, corresponding to 10^7 K. It matters only when use_superad_reduction=.true.; the three PPISN pulse cases set it explicitly to 7d0 in inlist_ppisn.

WHY

Confine this envelope stabilization to the intended temperature range without introducing a solver-iteration-dependent switch.

PRIMARY CODE

star/private/turb_support.f90; star/defaults/controls.defaults; PPISN inlist_ppisn files.

Bug Fixes

Corrections to existing MESA conservation, state handling, boundaries, solver scaling, and reported quantities.

11. Conservative velocity remap during split and merge

BUG FIX — U_FLAG / AMR / CONSERVATION

CHANGELOG: NO

PREVIOUS IMPLEMENTATION

A merge preserved kinetic-energy magnitude rather than momentum:

$$K_{\text{old}} = \frac{1}{2} dm_1 u_1^2 + \frac{1}{2} dm_2 u_2^2, \quad u_0 = \text{sign}(u_1) \sqrt{\frac{2K_{\text{old}}}{dm_1 + dm_2}}.$$

In general $(dm_1 + dm_2)u_0 \neq dm_1 u_1 + dm_2 u_2$. A split reconstructed u^2 ,

$$u_R^2 = u^2 + \frac{1}{4} \nabla(u^2) \Delta r, \quad u_L^2 = \frac{dm u^2 - dm_R u_R^2}{dm_L},$$

then assigned the parent's sign. It preserved kinetic energy but not child momentum, and a large child contrast could drive the colder child's internal energy below the original stencil minimum.

CURRENT IMPLEMENTATION

The merge uses the momentum projection

$$u_0 = \frac{dm_1 u_1 + dm_2 u_2}{dm_1 + dm_2}, \quad \Delta K = \frac{1}{2} \frac{dm_1 dm_2}{dm_1 + dm_2} (u_1 - u_2)^2,$$

and deposits ΔK in internal energy. A split reconstructs velocity,

$$u_R = u + \frac{dm_L}{dm} \Delta u, \quad u_L = u - \frac{dm_R}{dm} \Delta u,$$

so $dm_R u_R + dm_L u_L = dm u$. The newly resolved specific kinetic energy

$$\Delta e_{\text{KE}} = \frac{dm_R dm_L}{2dm^2} (\Delta u)^2$$

is subtracted from both child internal energies. If it would create a new stencil minimum, the slope is reduced by $\theta = \sqrt{\Delta e_{\text{max}} / \Delta e_{\text{KE}}}$.

CONTROL / ACTIVATION

Automatic for `u_flag` split/merge AMR; no new control. Activation follows the existing `use_split_merge_amr` path.

WHY

Conserve radial momentum and internal-plus-resolved kinetic energy while preventing remeshing from creating an EOS-invalid cold child.

PRIMARY CODE

star/private/adjust_mesh_split_merge.f90.

12. Surface-cell mass floor in split/merge AMR

BUG FIX — U_FLAG / AMR

CHANGELOG: NO

PREVIOUS IMPLEMENTATION

The standard remesher used `min_surface_cell_dq`, but split/merge AMR tested zone 1 only against its global AMR floor,

$$dq_{\text{min},1}^{\text{old}} = \text{split_merge_amr_dq_min}.$$

It could therefore split the surface cell below the separately configured surface floor.

CURRENT IMPLEMENTATION

Zone 1 now uses

$$dq_{\text{min},1} = \max(\text{split_merge_amr_dq_min}, \text{min_surface_cell_dq}).$$

A conservative mandatory merge pass before and after ordinary AMR merges zones 1 and 2 until the surface floor is restored.

CONTROL / ACTIVATION

Existing `min_surface_cell_dq=1d-99`. No new toggle; split/merge AMR now honors the configured value.

WHY

Prevent extremely small dynamic surface cells and repeated boundary split/merge behavior.

PRIMARY CODE

star/private/adjust_mesh_split_merge.f90.

13. Minimum radial spacing in split/merge AMR

BUG FIX — AMR / MESH GEOMETRY

CHANGELOG: NO

PREVIOUS IMPLEMENTATION

The standard remesher enforced `mesh_min_dlnR`; the split/merge path had no corresponding pre-split child check and no mandatory repair pass. It could retain or create adjacent radii separated by only a few floating-point increments.

CURRENT IMPLEMENTATION

For $r_m = (r_o + r_i)/2$, a proposed split must satisfy

$$\min[\ln(r_o/r_m), \ln(r_m/r_i)] \geq 2 \text{mesh_min_dlnR}.$$

When repair is requested, cells with $\ln(r_o/r_i) < \text{mesh_min_dlnR}$ are mandatorily merged before ordinary AMR operations.

CONTROL / ACTIVATION

Existing `mesh_min_dlnR=1d-9`. A positive value limits proposed children. Existing `merge_if_dlnR_too_small=.false.`; set it to `.true.` to enable forced repairs. A true center is exempt because $\ln 0$ is undefined.

WHY

Make split/merge AMR obey the same radial-geometry validity floor as the standard mesh path.

PRIMARY CODE

star/private/adjust_mesh_split_merge.f90.

14. Conservative velocity transport during mass loss

BUG FIX — U_FLAG / V_FLAG / ADJUST_MASS

CHANGELOG: NO

PREVIOUS IMPLEMENTATION

`do_adjust_mass` revised dq , dm , composition, rotation, and radius, but it did not interpolate or remap u or v . Velocity values remained attached to zone indices while the control mass underneath those indices changed. Consequently neither

$$P = \sum_i dm_i u_i \quad \text{nor} \quad K = \frac{1}{2} \sum_i dm_i u_i^2$$

was transported by the same mass overlaps used for the other conserved fields.

CURRENT IMPLEMENTATION

For overlap

$$\Delta m_{ij} = \max[0, \min(b_{i+1}^{\text{old}}, b_{j+1}^{\text{new}}) - \max(b_i^{\text{old}}, b_j^{\text{new}})],$$

the new cell receives

$$p_j = \sum_i \Delta m_{ij} u_i, \quad u_j = p_j / dm_j, \quad \Delta K_j = \frac{1}{2} \sum_i \Delta m_{ij} u_i^2 - \frac{p_j^2}{2dm_j} \geq 0.$$

$\Delta K_j / (dm_j \Delta t)$ is restored once as a local non-leaking source. The staggered path uses $dm_1^v = dm_1/2$ and $dm_k^v = (dm_{k-1} + dm_k)/2$.

CONTROL / ACTIVATION

Automatic for hydrodynamic mass loss when `u_flag` or `v_flag` is active; no new control. Accretion is unchanged because incoming momentum requires a separately defined boundary state.

WHY

Preserve momentum and total energy when hydrodynamic mass removal changes the mass grid.

PRIMARY CODE

star/private/adjust_mass.f90; star/private/eps_mdot.f90.

15. Positive optical depth after a surface cut

BUG FIX — SURFACE / OPTICAL DEPTH / AMR

CHANGELOG: YES

PREVIOUS IMPLEMENTATION

Whenever the momentum boundary was active, surface removal installed

$$f_{\tau, \text{new}} = f_{\tau, \text{old}} \frac{\tau_{\text{eff}}}{\tau_1} = \frac{\tau_{\text{eff}}}{\tau_{\text{base}}},$$

$$\tau_{\text{eff}} = \kappa \left[\frac{P_{\text{EOS}}}{g} - f_{P, \text{extra}} \frac{L/M}{6\pi c G} \right]$$

without checking its sign. A super-Eddington retained cell could yield $\tau_{\text{eff}} \leq 0$, after which metric zoning evaluated $\ln \tau$ on invalid data.

CURRENT IMPLEMENTATION

The estimate replaces `tau_factor` only when finite and positive; otherwise the incoming factor is retained. Metric zoning likewise uses its $\ln \tau$ component only when all required optical-depth boundaries and `tau_center` are finite and positive. Invalid optical depth disables only that component for the current remesh call.

CONTROL / ACTIVATION

Automatic in the surface-removal and metric-AMR paths; no new control. Radius metric zoning remains active when its weight is positive.

WHY

Prevent a nonphysical optical depth from causing invalid logarithms and runaway refinement while preserving independent radius zoning.

PRIMARY CODE

`star/private/remove_shells.f90`; `star/private/adjust_mesh_split_merge.f90`.

16. Photospheric luminosity used by effective temperature

BUG FIX — REPORTING / PHOTOSPHERE

CHANGELOG: YES

PREVIOUS IMPLEMENTATION

For an interior photosphere, MESA first computed

$$f = \frac{\tau_{\text{phot}} - \tau_k}{\Delta \tau_k}, \quad L_{\text{int}} = L_k + f(L_{k+1} - L_k),$$

but then discarded it with `Lphot=max(1d0, L(1))`. Thus R_{phot} came from $\tau = \tau_{\text{base}}$ while L_{phot} came from the surface.

CURRENT IMPLEMENTATION

The floor is applied to the interpolated value,

$$L_{\text{phot}} = \max[1 \text{ erg s}^{-1}, L_{\text{int}}], \quad T_{\text{eff}} = \left(\frac{L_{\text{phot}}}{4\pi R_{\text{phot}}^2 \sigma} \right)^{1/4}.$$

The surface fallback and standard surface `log_L` column are unchanged.

CONTROL / ACTIVATION

Automatic when the photosphere is located inside the model; no new control.

WHY

Make photospheric radius, luminosity, and effective temperature refer to the same optical-depth location.

PRIMARY CODE

`star/private/star_utils.f90`.

17. Post-hydro temperature convergence floor

BUG FIX — HYDRO / CONVERGENCE

CHANGELOG: YES

PREVIOUS IMPLEMENTATION

The hydro matrix used the configurable lower bound, but the accepted-model check independently rejected

$$\log_{10} T < 1$$

with a hard-coded value. Lowering `hydro_mtx_min_allowed_logT` therefore did not lower the final convergence floor.

CURRENT IMPLEMENTATION

The post-solve check now rejects

$$\log_{10} T < \text{hydro_mtx_min_allowed_logT},$$

the same limit used by the hydro matrix.

CONTROL / ACTIVATION

Existing `hydro_mtx_min_allowed_logT=1d0`. No new toggle; the accepted-model check now honors the configured value.

WHY

Use one documented temperature limit consistently in the Newton solve and acceptance test.

PRIMARY CODE

`star/private/hydro_eqns.f90`.

18. Finite luminosity correction weights

BUG FIX — HYDRO / SOLVER

CHANGELOG: YES

PREVIOUS IMPLEMENTATION

The Newton correction weight was

$$w_L^{\text{old}}(k) = \frac{1}{0.1 L_{\text{start}}(1) + |L(k)|}.$$

At zero luminosity the denominator could vanish; a negative starting surface luminosity could also cancel the positive $|L(k)|$ term and make the scale nonpositive or singular.

CURRENT IMPLEMENTATION

The weight is

$$w_L(k) = \frac{1}{\max[1 \text{ erg s}^{-1}, 0.1 |L_{\text{start}}(1)| + |L(k)|]}.$$

CONTROL / ACTIVATION

Automatic when the existing `include_L_in_correction_limits=.true.`; no new control. Setting that existing control false omits luminosity from correction-size limits while retaining the luminosity equation and residual test.

WHY

Keep the correction norm finite and positive for zero or inward luminosity.

PRIMARY CODE

`star/private/solver_support.f90`.

19. Pole-free luminosity split in the dP_{rad}/dm equation

BUG FIX — THERMAL TRANSPORT / TDC / RTI

CHANGELOG: YES

PREVIOUS IMPLEMENTATION

On a face recognized as convective, the equation used

$$L_{\text{rad}} = L \frac{\nabla_T}{\nabla_{\text{rad}}}.$$

This is 0/0 when L and ∇_{rad} cross zero. The old branch also forced $L_{\text{rad}} = L$ for negative ∇_{rad} and tested final `mixing_type`; RTI could replace that label even while the MLT/TDC closure still supplied ∇_T and L_{conv} .

CURRENT IMPLEMENTATION

The coefficient is evaluated without dividing by luminosity or ∇_{rad} :

$$L_0 = \frac{16\pi c G m P_{\text{rad}}}{f_{\nabla} P_{\kappa}}, \quad L_{\text{rad}} = L_0 \nabla_T, \quad L_{\text{conv}} = L - L_{\text{rad}}.$$

The split tests `mlt_mixing_type`, so RTI chemical mixing cannot hide an otherwise active MLT/TDC thermal closure.

CONTROL / ACTIVATION

Existing `use_dPrad_dm_form_of_T_gradient_eqn=.false.`; set it to `.true.` to select this temperature equation. The corrected split is then automatic.

WHY

Remove a removable pole, retain counterflowing convection, and keep the temperature equation consistent with the closure that produced ∇_T .

PRIMARY CODE

`star/private/hydro_temperature.f90`.

20. Stress-free TDC surface traction

BUG FIX — TDC / U_FLAG / BOUNDARY

CHANGELOG: YES

PREVIOUS IMPLEMENTATION

For the missing exterior cell, velocity and radius wrappers returned $u_0 = 0$ and $r_0 = 0$; the radius was then replaced by 1 cm to avoid division by zero. The surface strain therefore contained

$$\left(\frac{u_0}{r_0} - \frac{u_1}{r_1} \right) \longrightarrow -\frac{u_1}{r_1},$$

creating a nonzero exterior eddy stress solely from the wrapper convention.

CURRENT IMPLEMENTATION

The outer TDC face is explicitly stress free for cell-centered hydro,

$$\chi_1 = 0, \quad U_{q,1} \propto \chi_1 - \chi_2 = -\chi_2.$$

The surface cell therefore retains the physical inner-face traction without an artificial exterior strain.

CONTROL / ACTIVATION

Automatic for `u_flag` TDC eddy viscosity; no new control. The term is active through the existing TDC/TDC_alpha_M settings.

WHY

Impose the intended stress-free boundary while retaining the interior turbulent force.

PRIMARY CODE

`star/private/tdc_hydro.f90`.

21. Consistent radius collocation in TDC eddy viscosity

BUG FIX — TDC / U_FLAG / EDDY VISCOSITY

CHANGELOG: YES

PREVIOUS IMPLEMENTATION

Three related operators used different radii or time levels. The χ strain used current iterate radii,

$$z_k^\chi = u_k / r_k,$$

the heating strain used time-centered $\bar{u}_k / \bar{r}_k$, and the force denominator used the time-centered cell midpoint $(\bar{r}_k + \bar{r}_{k+1})/2$. Because these were not one discrete operator, $\sum_k u_k F_k$ had no guaranteed nonpositive form.

CURRENT IMPLEMENTATION

For `u_flag`, all three use the fixed start-of-step midpoint

$$r_k^* = r_{\text{mid,start},k}, \quad z_k = u_k / r_k^*,$$

$$\chi_k = C_k(z_{k-1} - z_k), \quad F_k = \frac{4\pi(\chi_k - \chi_{k+1})}{r_k^*}, \quad \sum_k u_k F_k = -4\pi \sum_k C_k(z_{k-1} - z_k)^2 \leq 0$$

apart from boundary terms. Remeshing regenerates `rmid_start` from the final geometry; the velocity coupling remains block tridiagonal.

CONTROL / ACTIVATION

Automatic for `u_flag` TDC eddy viscosity; no new control. The `v_flag` discretization is unchanged.

WHY

Make viscous strain, heating, and force share one collocation and recover the expected dissipative work sign.

PRIMARY CODE

`star/private/tdc_hydro.f90`; `star/private/star_utils.f90`.

22. Compatible TDC eddy-viscous work in total energy

BUG FIX — TDC / ENERGY

CHANGELOG: YES

PREVIOUS IMPLEMENTATION

The energy source always contributed viscous heating $E_{q,k}$. Even when the active residual also contained dK/dt , it omitted the mechanical work of the momentum acceleration $U_{q,k}$:

$$S_{q,k}^{\text{old}} = E_{q,k}.$$

An accelerated cell could therefore pay for its kinetic-energy increase by losing internal energy.

CURRENT IMPLEMENTATION

When the residual contains radial kinetic energy,

$$S_{q,k}^u = E_{q,k} + \bar{u}_k U_{q,k}, \quad \bar{u}_k = \frac{u_k^{n+1} + u_k^n}{2}.$$

For staggered velocity,

$$S_{q,k}^v = E_{q,k} + \frac{1}{2}(\bar{v}_k U_{q,k} + \bar{v}_{k+1} U_{q,k+1}),$$

matching MESA's half-cell kinetic-energy quadrature. Internal-energy and time-centered $P d(1/\rho)$ forms continue to use E_q alone.

CONTROL / ACTIVATION

Automatic; no new control. The branch follows whether the selected energy residual actually contains dK/dt , avoiding both omission and double counting.

WHY

Make the momentum and total-energy equations exchange the same kinetic energy.

PRIMARY CODE

`star/private/hydro_energy.f90`; `star/private/tdc_hydro.f90`.

23. RTI state restoration and conservative AMR remap

BUG FIX — RTI / AMR / RELAXATION

CHANGELOG: YES

PREVIOUS IMPLEMENTATION

`relax_to_star_cut` could return with RTI disabled even when it was enabled on entry. During a split, an initial nonnegative interpolation was followed by an assignment that copied the already modified outer-child value into the inner child; in general,

$$dm_R \alpha_R + dm_L \alpha_L \neq dm \alpha_{\text{parent}}.$$

The behind-shock ramp also evaluated $(m_{\text{shock}} - m)/\Delta m$ at the default $\Delta m = 0$ and could become negative ahead of the shock.

CURRENT IMPLEMENTATION

The cut restores the incoming RTI flag and initializes the reconstructed retained model with zeroed α_{RTI} . Split children use

$$\alpha_R = \alpha + \frac{dm_L}{dm} \Delta \alpha, \quad \alpha_L = \alpha - \frac{dm_R}{dm} \Delta \alpha,$$

so $dm_R \alpha_R + dm_L \alpha_L = dm \alpha$. The slope is bounded to keep both children nonnegative and inside the original stencil. A zero-width ramp is skipped; a positive-width ramp is bounded below by zero.

CONTROL / ACTIVATION

Automatic when the existing `RTI_flag` is active; no new general MESA control. The PPISN test-local selector is documented in item 33.

WHY

Keep RTI state restart/cut safe and prevent remeshing or shock localization from creating nonconservative or negative α_{RTI} .

PRIMARY CODE

star/private/remove_shells.f90; star/private/adjust_mesh_split_merge.f90; star/private/mix_info.f90.

24. Compatible energy for RTI momentum diffusion

BUG FIX — RTI / U_FLAG / ENERGY

CHANGELOG: YES

PREVIOUS IMPLEMENTATION

RTI velocity diffusion supplied a momentum acceleration but the total-energy equation contained neither its midpoint work nor the heat associated with the lost resolved velocity variance. The source was effectively

$$S_{\text{RTI}}^{\text{old}} = 0$$

for that momentum term, so a cell accelerated by smoothing could cool internally.

CURRENT IMPLEMENTATION

Momentum and energy share

$$F_k = \sigma_k(u_{k-1} - u_k) - \sigma_{k+1}(u_k - u_{k+1}),$$

and total energy receives

$$S_k = \frac{\bar{u}_k F_k + Q_k}{dm_k}, \quad \sum_k (\bar{u}_k F_k + Q_k) = 0.$$

Q_k deposits the symmetric interface dissipation as heat; the existing RTI internal-energy diffusion remains a separate flux divergence.

CONTROL / ACTIVATION

Automatic for the existing cell-centered RTI momentum-diffusion path (u_flag with RTI_flag); no new control.

WHY

Convert RTI-smoothed resolved kinetic energy to heat locally while preserving global total energy.

PRIMARY CODE

star/private/hydro_riemann.f90; star/private/hydro_energy.f90.

25. Fe-core infall reporting for both velocity formulations

BUG FIX — REPORTING / U_FLAG / V_FLAG

CHANGELOG: YES

PREVIOUS IMPLEMENTATION

The surrounding report code selected `velocity => s%u` for `u_flag` and `velocity => s%v` for `v_flag`, but the peak-infall branch bypassed that pointer:

$$v_{\text{infall,Fe}}^{\text{old}} = -\min[s\%v(k_{\text{Fe}} : n_z)].$$

It therefore read the staggered array even for cell-centered hydro.

CURRENT IMPLEMENTATION

The diagnostic now uses the selected active array,

$$v_{\text{infall,Fe}} = -\min[\text{velocity}(k_{\text{Fe}} : n_z)].$$

CONTROL / ACTIVATION

Automatic when `report_max_infall_inside_fe_core=.true.` and either velocity formulation is active; no new control.

WHY

Report the Fe-core infall speed from the velocity representation actually being solved.

PRIMARY CODE

star/private/report.f90.

26. Correct density-inversion scale in superadiabatic reduction

BUG FIX — MLT / TDC

CHANGELOG: YES

PREVIOUS IMPLEMENTATION

The density-inversion contribution read `superad_reduction_Gamma_inv_scale` into a local value but multiplied its response by `superad_reduction_Gamma_limit_scale`. The inversion-scale control therefore had no effect when the two values differed.

CURRENT IMPLEMENTATION

For $x = L_{\text{rad}}/L_{\text{inv}} - 1$, the inversion contribution now uses its own scale,

$$\Gamma_{\text{inv}} = \text{superad_reduction_Gamma_inv_scale} \begin{cases} x^2/2, & 0 < x < 1, \\ x - 1/2, & x \geq 1. \end{cases}$$

The independent Eddington-limit contribution continues to use `superad_reduction_Gamma_limit_scale`.

CONTROL / ACTIVATION

Existing controls only; defaults are `superad_reduction_Gamma_limit_scale=5d0` and `superad_reduction_Gamma_inv_scale=5d0`. Equal values are unchanged; different values now act independently.

WHY

Restore the published two-scale algorithm and make the density-inversion control operational.

PRIMARY CODE

`star/private/turb_support.f90`.

27. Preserve turnover-limiter state during photo loading

BUG FIX — MLT / TDC / PHOTOS

CHANGELOG: YES

PREVIOUS IMPLEMENTATION

Current-format photos restored `superad_reduction_factor`, but `finish_load_model` could reset it to one while rebuilding derived quantities if `run_star_extras` had not yet restored the controls that enable superadiabatic reduction. The saved state was therefore lost before `extras_startup`.

CURRENT IMPLEMENTATION

When a photo supplied the factor, `finish_load_model` now treats it as authoritative and holds it independently of the controls active during load reconstruction. Outside that reconstruction, the factor is held only for an active turnover limiter until accepted start state is available. A run that leaves reduction disabled resets the factor on its first real evolution step.

CONTROL / ACTIVATION

Automatic for current-format photo restarts; no new control. Model-file loading is unchanged because it does not set `have_superad_reduction_factor`.

WHY

Make photo restarts reproduce the accepted turnover-limiter state even when phase-dependent controls are restored later by extras.

PRIMARY CODE

`star/private/photo_in.f90`; `star/private/turb_support.f90`.

Development and Test Changes

Test-driver and development-case changes. Test-local `x_ctrl` selectors are stated here; they are not new general MESA controls.

28. Phase-dependent PPISN timestep workflow

DEVELOPMENT/TEST — PPISN

CHANGELOG: NO

PREVIOUS IMPLEMENTATION

The pulse driver had a Riemann-hydro timestep ceiling only before it recognized an active pulse. The post-first-pulse `v_flag` ceiling lived in a separate `inlist_after_first_pulse`, so later phase `inlist` reads could overwrite it. The driver did not expose an independent active-pulse ceiling.

CURRENT IMPLEMENTATION

The driver applies all three phases directly. While `u_flag` is active it uses `x_ctrl(10)` before pulse onset and `x_ctrl(22)` after onset. While `u_flag` is off, it leaves the timestep unlimited before the first Riemann phase and caps it at 0.5 yr after that phase. Pulse time, dynamical time, and relaxation counters remain restart state. The obsolete after-first-pulse `inlist` is removed.

CONTROL / ACTIVATION

The PPISN `inlists` set `x_ctrl(10)=1d4 s` and `x_ctrl(22)=1d4 s`. A nonpositive `x_ctrl(22)` disables only the active-pulse ceiling. `restore_mesh_on_retry=.false.` in both hydro phase `inlists`; this workflow does not restore the pre-remesh grid. `split_merge_amr_avoid_repeated_remesh=.true.` independently prevents a cell from being split or merged repeatedly within one AMR pass.

WHY

Keep the timestep policy tied to the stored physical phase rather than to the order in which auxiliary `inlists` are reread.

PRIMARY CODE

`star/test_suite/ppisn/src/run_star_extras.f90`; the two PPISN development-case copies.

29. Direct PPISN ejecta removal

DEVELOPMENT/TEST — PPISN / EJECTA

CHANGELOG: NO

PREVIOUS IMPLEMENTATION

The older selector included a sound-speed gate and used `k_keep` inconsistently as either the last removed or first retained cell. A layer could stop being removed when $u/c_s < 1$ even while it remained strongly above escape speed. Removed-energy accounting did not consistently use the EOS internal energy stored in `s%energy`.

CURRENT IMPLEMENTATION

Starting at the surface, a contiguous cell is eligible only if it passes the configured extent gate and

$$u_i > f_{\text{esc}} \sqrt{\frac{2G_i m_i}{r_i}}, \quad e_{\text{tot},i} = \frac{u_i^2}{2} + e_i - \frac{G_i m_i}{r_i} > 0.$$

`k_keep` is always the first retained cell; ejecta sums use $1 : k_{\text{keep}} - 1$, and the direct mode prunes exactly those cells. Internal-energy accounting uses the EOS value in `s%energy`. The unused smooth mass-loss mode and its inverse-timescale control were removed.

CONTROL / ACTIVATION

Test-local `x_integer_ctrl(2)` accepts only 0 (disabled) or 1 (direct). `x_ctrl(18)` is the minimum radius in $R_\odot$ unless `x_ctrl(21)>0`, in which case `x_ctrl(21)` is the maximum local escape speed in km s^{-1} . `x_ctrl(19)` is f_{esc} . All three pulse cases select direct removal with `x_ctrl(18)=1d4`, `x_ctrl(19)=4d0`, and `x_ctrl(21)=0d0`.

WHY

Select genuinely unbound outward material and make retained indexing, removed mass, and removed energy use one definition.

PRIMARY CODE

`star/test_suite/ppisn/src/run_star_extras.f90`; the two PPISN development-case copies.

30. Post-cut boundary and relaxation state

DEVELOPMENT/TEST — PPISN / BOUNDARY

CHANGELOG: NO

PREVIOUS IMPLEMENTATION

After a direct computational prune, the emergency fixed surface temperature/velocity and optical-depth adjustment existed only in the immediate call path. A later control reread or photo restart could replace them while hydro was still active. The same emergency boundary state could also leak into the later physical `star_relax_to_star_cut` operation.

CURRENT IMPLEMENTATION

Direct removal sets

$$T_{\text{surf, fixed}} = T(k_{\text{keep}}), \quad v_{\text{surf, fixed}} = \max \left[u(k_{\text{keep}}), 2\sqrt{\frac{2GM(k_{\text{keep}})}{R_{\text{limit}}}} \right]$$

when the radius gate is active, subject to the configured emergency cap. The surface-removal routine updates f_r from the first retained cell. Temperature, velocity, and f_r are stored in restart data and reapplied after inlist reads or photos while `u_flag` remains active. Both the fixed-velocity and momentum flags remain true, so fixed velocity supplies momentum while pressure and temperature stay face-attached. Before the final physical cut, the latch is cleared, `tau_factor=1`, fixed velocity is disabled, the normal atmosphere is restored, and RTI is disabled before `u_flag`.

CONTROL / ACTIVATION

Automatic inside direct mode from item 29. Test-local `x_ctrl(17)=5d4 km s-1` caps the emergency fixed velocity. The persistent boundary latch is internal restart state, not a general MESA control.

WHY

Separate the temporary extended-layer prune boundary from the final hydrostatic cut and make both paths restart deterministic.

PRIMARY CODE

`star/test_suite/ppisn/src/run_star_extras.f90`; the two PPISN development-case copies.

31. Ideal-EOS electron correction for PPISN opacity

DEVELOPMENT/TEST — PPISN / OPACITY

CHANGELOG: NO

PREVIOUS IMPLEMENTATION

The ideal-EOS coverage fallback supplies a nominal electron number density of `1d-20`, so its opacity input is

$$\ln f_{e,I} = \ln \left(\frac{10^{-20}}{N_A \rho} \right).$$

When that component entered an EOS blend in dilute hot ejecta, the blended `lnfree_e` could artificially suppress Compton opacity.

CURRENT IMPLEMENTATION

Above `logT=3.7`, the case-local opacity hook replaces only the ideal-EOS contribution. With ideal fraction f_I and electron fraction Y_e ,

$$\Delta_I = \ln Y_e - \ln f_{e,I}, \quad \ln f_{e,\text{kap}} = \ln f_e + f_I \Delta_I.$$

Its density and temperature derivatives include both $f_I \Delta_I$ and the derivatives of f_I . The EOS thermodynamic state and stored `lnfree_e` are unchanged; only the arguments passed to `kap_get` are corrected.

CONTROL / ACTIVATION

Case-local and automatic when `use_other_kap=.true.`, k identifies a real zone, `logT>3.7`, $f_I > 0$, and $Y_e > 0$. There is no new MESA control.

WHY

Keep the ideal-gas pressure fallback from behaving like a neutral-electron opacity model in hot dilute PPISN layers.

PRIMARY CODE

The three PPISN `src/run_star_extras.f90` files; `notes/other_kap_hot_ideal_free_e_fix.md`.

32. Smooth low-temperature opacity continuation

DEVELOPMENT/TEST — PPISN / OPACITY

CHANGELOG: NO

PREVIOUS IMPLEMENTATION

Below the tabulated lower boundary, repeated PPISN retries could return to the same invalid opacity request near $\log T = 2.7$; reducing the timestep did not move the trial state back into the table.

CURRENT IMPLEMENTATION

Let $x = \log_{10} T$, $x_0 = 2.7001$, $x_1 = 2.8001$, $u = (x - x_0)/(x_1 - x_0)$, and $w = 3u^2 - 2u^3$. In logarithmic opacity,

$$\ell_{\text{ext}} = \begin{cases} \ell(x_0), & x \leq x_0, \\ (1 - w)\ell(x_0) + w\ell(x), & x_0 < x < x_1, \\ \ell(x), & x \geq x_1. \end{cases}$$

The hook returns the matching density, temperature, and composition derivatives and blends `kap_fracs` with the same w . The boundary opacity is reevaluated at the current density and composition, so this is not a constant opacity floor. The existing escaping-surface electron-scattering override retains precedence.

CONTROL / ACTIVATION

Case-local and automatic when `use_other_kap=.true.`; no new MESA control or restart state. Above $\log T = 2.8001$, the ordinary opacity call is unchanged.

WHY

Continue the table edge with a continuous first temperature derivative while leaving core MESA opacity behavior untouched.

PRIMARY CODE

The three PPISN `src/run_star_extras.f90` files; `notes/other_kap_hot_ideal_free_e_fix.md`.

33. RTI lifecycle and diagnostics in the PPISN case

DEVELOPMENT/TEST — PPISN / RTI

CHANGELOG: NO

PREVIOUS IMPLEMENTATION

The PPISN driver did not own the complete variable-set transition. RTI could be enabled/disabled by control rereads without a defined ordering relative to `u_flag`; a restart could reinitialize α_{RTI} , and a cut could disable hydro before RTI. Output did not expose the full source–diffusion–momentum–energy budget.

CURRENT IMPLEMENTATION

The transition is explicit: hydro off implies RTI off; hydro on first selects `u_flag`, then optionally enables RTI; a new pulse zeros α_{RTI} ; an already-active restart preserves it; a cut disables RTI before disabling `u_flag`. Profiles/PGSTAR include α_{RTI} , growth/decay sources, diffusion, face kick, momentum/energy terms, and Riemann-shock diagnostics.

CONTROL / ACTIVATION

Test-local `x_logical_ctrl(3)` selects RTI during Riemann-hydro phases. `.false.` keeps TDC and its independently configured eddy viscosity without RTI; `.true.` enables the lifecycle above. The PPISN test and both pulse development cases currently set it to `.false.`

WHY

Make RTI state transitions reproducible and expose enough diagnostics to separate RTI energetics from TDC eddy viscosity.

PRIMARY CODE

`star/test_suite/ppisn/src/run_star_extras.f90`; the two PPISN development-case copies and diagnostic lists.

34. Added Noh Riemann development case

DEVELOPMENT/TEST — NOH / U_FLAG

CHANGELOG: NO

PREVIOUS IMPLEMENTATION

No focused current-format TDC development case exercised the cell-centered HLLC solver against the spherical Noh solution. The suite therefore had no automated comparison with the $\gamma = 5/3$, unit-inflow postshock state

$$\rho_2 = 64\rho_0, \quad P_2 = \frac{64}{3}\rho_0 u_0^2, \quad e_2 = \frac{1}{2}u_0^2,$$

or a cell-by-cell accounting of the wall-heating error.

CURRENT IMPLEMENTATION

`dev_TDC_noh_riemann` creates a fresh gamma-law model, runs cell-centered Riemann hydro, checks shock/wall behavior, and records

$$\Delta U_k = dm_k(e_k - e_{k,\text{start}}), \quad \Delta K_k = \Delta t \, dm_k \dot{K}_k, \quad \Delta W_k = \Delta t \, dm_k \dot{W}_k,$$

with $B_k = \Delta U_k + \Delta K_k + \Delta W_k$. Plotting scripts compare the numerical postshock state and integrated closure with the analytic solution.

CONTROL / ACTIVATION

Case-local setup: `constant_L=.true.`, `include_L_in_correction_limits=.false.`, `hydro_mtx_min_allowed_logT=-99`, and test hooks for $P = (\gamma-1)\rho e$, zero gravity, and constant opacity. These are benchmark settings, not new controls beyond item 6.

WHY

Isolate Riemann energy closure and the spherical Noh wall-heating error in a reproducible current-format case.

PRIMARY CODE

`star/dev_cases_test_TDC/dev_TDC_noh_riemann`.

35. Added Sedov $\rho \propto r^{-1}$ development case

DEVELOPMENT/TEST — SEDOV / AMR

CHANGELOG: NO

PREVIOUS IMPLEMENTATION

The current TDC development suite had no focused Sedov case for $\rho \propto r^{-1}$ and no direct comparison of the AMR shock trajectory with the similarity scaling

$$r_s(t) \propto \left(\frac{Et^2}{\rho_0} \right)^{1/(5-\omega)}, \quad \omega = 1.$$

Broad stellar-evolution cases could not isolate shock-position or mesh errors from the stellar physics.

CURRENT IMPLEMENTATION

`dev_TDC_sedov_omega_1` generates a fresh gamma-law model, supplies standard and high-resolution runs, records mesh/shock diagnostics, and compares density, pressure, velocity, energy, and $r_s(t)$ with the Kamm–Timmes Sedov solution.

CONTROL / ACTIVATION

Case-local setup mirrors the idealized closure in item 34: `constant_L=.true.`, luminosity excluded from correction-size limits, relaxed gamma-law temperature bound, zero gravity, and no MLT. Resolution and timestep variants are in the case inlists/run scripts.

WHY

Validate shock position, postshock structure, total energy, and AMR behavior against a known similarity solution.

PRIMARY CODE

`star/dev_cases_test_TDC/dev_TDC_sedov_omega_1`.

36. Added current-format weak-shock pulse case

DEVELOPMENT/TEST — TDC / WEAK SHOCKS

CHANGELOG: NO

PREVIOUS IMPLEMENTATION

The MESA IV experiment used removed inlist/extras interfaces and legacy fixed-luminosity surface controls. Its intended boundary was nevertheless precise:

$$T_{\text{surf}}^4 = \frac{L_{\text{fixed}}}{4\pi\sigma R_{\text{surf}}^2}, \quad \frac{\partial \ln T_{\text{surf}}}{\partial L} = 0, \quad \frac{\partial \ln T_{\text{surf}}}{\partial \ln R} = -\frac{1}{2}.$$

The legacy AMR pair `MaxLong=1.1`, `MaxShort=1.1` also allowed a freshly split cell to be immediately merge eligible because their product was below two.

CURRENT IMPLEMENTATION

`dev_TDC_weak_shocks` prepares a $0.8 M_\odot$ white-dwarf envelope, removes the core, settles with hydro, injects localized energy, and follows pulse breakout with TDC. A case-local `other_surface_PT` hook restores the exact blackbody closure and derivatives above; L_{fixed} is restart data and is reset after direct removal. The inlists retain `MaxLong=1.1` but use `MaxShort=2.5` for hysteresis.

CONTROL / ACTIVATION

No new general control. The boundary is installed by this development case's `other_surface_PT` hook; its prepare/injection/finish inlists select the existing compression or fixed-velocity momentum boundary as appropriate.

WHY

Port the published weak-shock experiment without reintroducing obsolete global controls and remove an AMR merge-split ambiguity.

PRIMARY CODE

star/dev_cases_test_TDC/dev_TDC_weak_shocks.

37. Reorganized and synchronized PPISN models

DEVELOPMENT/TEST — PPISN / CASE ORGANIZATION

CHANGELOG: NO

PREVIOUS IMPLEMENTATION

The single dev_TDC_through_ppisn directory mixed starting-model construction with an older pulse workflow. There was no maintained binary-progenitor directory or distinct pulse case for the binary product, and the test-suite case had diverged from the development physics and source.

CURRENT IMPLEMENTATION

The workflow is separated into three development directories. `binary_ppisn_progenitor` evolves an initially $100 + 50 M_{\odot}$, 10-day binary and writes `MASS2accretor_final.mod`. `dev_TDC_ppisn_from_binary` loads that handoff and follows its pulses. `dev_TDC_single_star_ppisn` constructs an isolated $72.5 M_{\odot}$ helium star and follows its pulses. Both pulse development cases are open ended. The maintained test uses the single-star physics but stops 100 days after the first relaxation and has `max_model_number=15000`.

CONTROL / ACTIVATION

The three development `rn` and `re` scripts select their top-level inlists explicitly with `MESA_INLIST`; they do not load test-suite helpers or install `test_suite_extras`. The test-suite `rn` and `extras` retain that bookkeeping and verify the 100-day termination. Both pulse development sources are identical; the test copy differs only in test-suite integration and its success stop.

WHY

Give the isolated and binary-derived PPISN calculations explicit model handoffs while keeping one shared pulse implementation and a bounded regression test.

PRIMARY CODE

star/dev_cases_test_TDC/ppisn_models; star/test_suite/ppisn.

38. Aligned PPISN microphysics and convection inputs

DEVELOPMENT/TEST — PPISN / PHYSICS

CHANGELOG: NO

PREVIOUS IMPLEMENTATION

The progenitor, isolated pulse case, binary-product pulse case, and test used different networks, EOS coverage thresholds, and TDC choices. The binary progenitor also lacked the pulse models' helium-core overshoot rule.

CURRENT IMPLEMENTATION

All four PPISN directories select `approx21_cr60_plus_co56.net`, `mass_fraction_limit_for_Skye=1d-5`, the deBoer $^{12}\text{C}(\alpha, \gamma)^{16}\text{O}$ special rate, `mixing_length_alpha=2d0`, `MLT_option='TDC'`, `include_mlt_corr_to_TDC=.false.`, turbulent energy in the energy equation, and `harmonic_dissipation_length_beta=1d0`. The pulse cases explicitly use `TDC_alpha_M=0d0`, zero thermohaline mixing, and the turnover-limited superadiabatic reduction with the $\log T = 7$ cutoff. The progenitor retains thermohaline mixing and its instantaneous reduction settings, and its helium-burning core uses $(f, f_0) = (0.01, 0.005)$. Both progenitor stars start without rotation, and `do_tidal_sync=.false.` disables the corresponding tidal torque.

CONTROL / ACTIVATION

These are explicit case inputs rather than new controls. The progenitor inherits the MESA default `TDC_alpha_M=0d0`. Wind prescriptions and phase-specific opacity, velocity, and ejecta controls remain intentionally different.

WHY

Keep handoff models on a consistent nuclear, EOS, and local-convection basis without copying pulse-only hydrodynamic treatments into binary evolution.

PRIMARY CODE

The four PPISN `inlist_ppisn` or shared progenitor inlists; `star/test_suite/20M_pre_ms_to_core_collapse/inlist_common` for the common network/EOS reference.

39. Replaced legacy broad TDC evolution cases

DEVELOPMENT/TEST — CASE ORGANIZATION

CHANGELOG: NO

PREVIOUS IMPLEMENTATION

Four broad directories duplicated old interfaces and end-to-end stellar workflows:

- `dev_TDC_to_cc_12` and `dev_TDC_to_cc_80`: generic evolution toward core collapse;
- `dev_TDC_to_pisn_200`: generic evolution toward PISN;
- `dev_TDC_to_ppisn_100`: generic evolution toward PPISN.

They used saved legacy models, obsolete inlist names/interfaces, and largely duplicated histories, profiles, and extras code; none isolated a single conservation or shock benchmark.

CURRENT IMPLEMENTATION

Those four directories are deleted. `dev_TDC_noh_riemann`, `dev_TDC_sedov_omega_1`, and `dev_TDC_weak_shocks` provide focused hydro validation, while the maintained PPISN test and the three directories under `star/dev_cases_test_TDC/ppisn_models` carry the progenitor and pulsational workflows.

CONTROL / ACTIVATION

No control. This is a development-suite replacement: targeted cases are selected by their own `rn` scripts and inlists.

WHY

Replace duplicated obsolete infrastructure with cases that each have a specific analytic, conservation, or pulsational validation purpose.

PRIMARY CODE

`star/dev_cases_test_TDC`.